Advanced Research Project

Zurich University of Applied Sciences

School of Management and Law

News-Driven Electricity Trading: Fine-Tuned NLP for Signal Generation in German Electricity Markets

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Management Summary

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1 Introduction

1.1 Background

~~The evolution of electricity systems has been characterised by the development of energy sources, with periods of both stagnation and rapid advancement. In recent times, there has been a resurgence in the adoption of renewable energy sources, marking a modern renaissance in this field. The 1970s oil crisis led to accelerated policy and technological shifts, which resulted in the revival of non-fossil energy sources. Today's power systems increasingly rely on a diverse mix of energy sources, including renewables like wind and solar power, along with established hydropower. This is reshaping the electricity mix towards cleaner sources (Lefort & Simic, 2024).~~

~~The utilisation of electricity produced by diverse sources can be facilitated through the acquisition in the capital markets. Electricity trading sits at the intersection of physical delivery and financial risk transfer: wholesale power is exchanged bilaterally and on organised markets, with balancing needs shaping how the market operates. A notable issue is the relative immaturity of electricity as a commodity, in contrast to other well-established products such as gold, coffee, or oil, with its ability to be stockpiled. Electricity is inherently subject to immediate consumption, which poses distinct challenges for its storage and distribution (Edwards, 2023, p. 2).~~

~~The exchange of power is conducted through two primary market mechanisms: day-ahead auctions and intraday continuous markets, which facilitate near-term scheduling. Additionally, the use of forwards and futures contracts enables the hedging of power requirements months to years in advance (Edwards, 2023, p. 2). Power producers agree to sell electricity to buyers at pre-agreed terms via purchase agreements or long-term contracts, often structured physically or financially. These are used for price certainty and financing (Edwards, 2023, p. 4).~~

NLP has the capacity to transform unstructured energy news (policy moves, outages, weather commentary) into structured signals that complement historical prices, thus facilitating effective forecasting and trading of electricity. It is evident that benchmarks on Australia's NSW market demonstrate the capacity to integrate price history, meteorological data and curated market news in order to predict next-day (48 half-hour steps) electricity prices (Bi et al., 2025, p. 1). In energy specifically, combining BERT-

based news features with gas price series has achieved >70% trend-direction accuracy, underscoring that sector news carries predictive content (Guan et al., 2022, p. 3573). Bai et al. (2024, p. 1) demonstrate that day-ahead forecasts benefit from the incorporation of textual features, including word frequencies, public sentiments, topic distributions, and word embeddings. These text features capture social events, including pandemics, politics, international conflicts and transportation, that influence electricity systems. This reinforces the feasibility of using news to enhance market decisions.

1.2 Research Question

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1.3 Objective and Significance

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1.4 Structure of the Research Paper

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2 Theoretical Foundations

2.1 Fundamentals of the Electricity Market

2.1.1 General Introduction to the Electricity Markets

The evolution of electricity systems has been characterised by the development of energy sources, with periods of both stagnation and rapid advancement. In recent times, there has been a resurgence in the adoption of renewable energy sources, marking a modern renaissance in this field. The 1970s oil crisis led to accelerated policy and technological shifts, which resulted in the revival of non-fossil energy sources. Today's power systems increasingly rely on a diverse mix of energy sources, including renewables like wind and solar power, along with established hydropower (Lefort & Simic, 2024). The underlying principle of renewable sources is that they are unlimited and not finite, in contrast to fossil fuels, which are utilised for conventional energy generation (Bundesnetzagentur, 2025a). As demonstrated by Statista (2025), the German market has experienced a notable increase in renewable electricity production, with wind power witnessing a 36% growth and solar power recording a substantial 64% increase from 2018 to 2025. This expansion plays an important role in reducing emissions from electricity generation. However, as Dannecker (2015, p. 17) illustrates in the German context, it also brings certain challenges and unintended side effects. Wind and solar energy production is highly dependent on weather conditions, resulting in inconsistent output and impacting electricity prices, which leads to increased volatility. Countries that want to rely heavily on renewables, such as Germany, are particularly affected by these fluctuating energy prices. A more recent example of this phenomenon is the recent blackout that occurred in Spain this year, which was attributed to an excess of renewable energy generation and a deficit of conventional energy (CNN, 2025). In this context, Dannecker (2015, p. 20) highlights the need for more advanced forecasting methods for renewable electricity, ensuring that production can be better aligned with demand and helping to stabilise the increased volatility observed in modern power markets.

2.1.2 Participants of the Electricity Markets

Electricity generated at power plants is fed into the high-voltage transmission grid, where it can be transported over long distances with minimal losses. Transmission system operators (TSOs) manage this high-voltage network by ensuring that electricity from

various generation facilities is safely integrated, routed, and delivered to regional distribution grids. At the European level, TSOs are coordinated by European Network of Transmission System Operators for Electricity (ENTSO-E), which harmonises grid operation, security standards, and cross-border electricity flows to support a reliable and interconnected power system (Bundesnetzagentur, 2025c). The transmission grid functions as the backbone of the electricity supply system, while the distribution grids pass electricity onward to households, firms, and smaller industrial consumers. Market participants purchase electricity on exchanges such as the European Power Exchange (EPEX) to either supply customers, hedge their exposures, or optimise their own generation portfolios (Bundesnetzagentur, 2025b). A notable issue is the relative immaturity of electricity as a commodity, in contrast to well-established products such as gold, coffee, or oil, which can be stored easily. Electricity, by contrast, must be consumed immediately, creating distinct challenges for its storage, pricing, and distribution (Edwards, 2023, p. 2).

2.1.3 Instruments in the markets (forwards/next-day auctions)

Power producers agree to sell electricity to buyers at pre-agreed terms via purchase agreements or long-term contracts, often structured physically or financially. These instruments are used to secure price certainty, facilitating financing, or, in some cases, to obtain exposure to market price fluctuations for speculative purposes (Edwards, 2023, p. 2). This can be achieved through bilateral over-the-counter (OTC) contracts or by trading through an intermediary such as EPEX, where transactions can be executed in the spot market or via derivatives (Edwards, 2023, p. 5). The present study focuses on spot and forward trading, while other contract forms are included for the sake of completeness, but fall outside the study's scope.

The two electricity markets under discussion in this research paper comprise the day-ahead auction and the intraday market, where they enable short-term balancing of supply and demand. The intraday market allows continuous trading, enabling participants to adjust positions in response to updated forecasts or unexpected changes in generation and consumption (EPEX Spot, 2025). A distinctive feature of the spot market is that purchased electricity must be consumed immediately, which requires direct physical delivery to the end user. As a result, the spot market is accessible primarily to companies that are able to either take delivery or utilise the electricity directly. It is important to recognise that the spot market is characterised by pronounced volatility, driven by a combination of internal

and external factors. Unexpected events, such as unplanned outages, can lead to sharp price increases as the market reacts to sudden shifts in supply (Edwards, 2023, p. 51). Conversely, electricity prices may also become negative – an uncommon occurrence in commodity markets – when excess generation exceeds system demand and available flexibility. Negative prices therefore function as a market signal to curb inflexible generation and, where possible, stimulate additional consumption, helping the system absorb temporary oversupply arising from high renewable feed-in and low demand (EPEX Spot, 2025).

Building on the characteristics of the spot market – where prices can fluctuate sharply due to volatility, scarcity, or even negative pricing – it is also important to consider how market participants manage these dynamics through longer-term instruments. In the day-ahead auction, bids and offers for each hourly delivery period of the following day are aggregated into a uniform market-clearing price once the auction closes at 11:00 UTC+1 (EPEX Spot, 2025). Beyond these hourly products, electricity is also traded using standardised delivery profiles. A base load contract represents all 24 hours of a given delivery period – such as a day, week, month, or year – and its price corresponds to the average of all hourly prices within that interval. A peak load contract, by contrast, covers only the hours with the highest typical demand, usually from 08:00 to 20:00 on working days, with its price reflecting the average of the hourly prices during this peak window, which are generally higher due to increased consumption (Müsgens & Bade, 2024, pp. 15–16).

Beyond these standardised delivery profiles, electricity markets also employ longer-term contractual instruments that allow price risks to be managed across extended horizons. Forward contracts are typically negotiated bilaterally over the counter and enable buyers and sellers to fix prices for future delivery periods, offering a high degree of flexibility with respect to maturity, volume and settlement structure (IEA, 2024, p. 62). Closely related to forwards, electricity swaps represent financially settled agreements that exchange fixed and floating price components and are widely used for risk management purposes. Complementing these bilateral instruments, futures contracts provide a standardised alternative that is traded on exchanges and cleared centrally, implying daily margining and reduced counterparty risk (IEA, 2024, p. 62, Mack, 2014, p. 52). Options further extend the set of available instruments by granting the right, but not the obligation, to buy or sell electricity or related underlyings, thereby allowing market participants to

limit downside risk while retaining exposure to favourable price movements (Mack, 2014, p. 40). Taken together, these instruments play an important role in electricity markets. Their practical use cases and trading motivations are examined in more detail in Section XY.

The coexistence of different delivery profiles, such as base and peak load, together with longer-term contractual instruments highlights the need to accommodate fluctuating electricity consumption and evolving risk exposures across time. While delivery profiles structure when electricity is supplied, forward and derivative products enable price risks associated with these profiles to be managed over extended horizons. Taken together, these instruments provide the foundation for organising trading activity in electricity markets and for translating physical system characteristics into market prices. Against this background, it becomes essential to consider the internal and external factors that influence price formation and shape the functioning of electricity markets more broadly.

2.1.4 Supply and Demand influences on the electricity markets

The existence of different delivery profiles, such as base and peak load, alongside the products traded in the spot and day-ahead markets, reflects the fact that electricity consumption does not remain constant throughout the day. These products allow market participants to address varying usage patterns and to handle the price movements that result from them. Against this background, it becomes essential to consider the supply and demand factors that shape price formation and determine how the electricity market functions more broadly. Although these factors are often conceptually separated, they are closely interlinked in practice, which means that a strict distinction is not always possible.

Previous sections have already highlighted demand-related factors such as base and peak load, which reflect the fact that electricity usage varies across the hours of the day. In addition, weekly cycles can also be observed, with noticeable differences between weekdays and weekends. These variations are largely driven by the recurring consumption behaviour of households and firms. Generators attempt to align production with these demand profiles as closely as possible, relying on forecasting models that are calibrated to expected usage patterns (Edwards, 2023, p. 135). Beyond daily and weekly patterns, electricity markets are also shaped by broader seasonal effects. The contrast between summer and winter plays a more pronounced role than in markets for storable commodities. For many of these commodities, storage creates a convenience yield – an

economic benefit of holding inventories – that often contributes to a smoother, upward-sloping term structure. Many energy markets, including gas and electricity, by contrast, cannot rely on economically viable large-scale storage, and their prices therefore tend to exhibit more distinct seasonal patterns, with higher levels in winter and lower levels in summer due to the corresponding shifts in demand. (Edwards, 2023, p. 77). Edwards (2023, p. 137) further shows a strong correlation between temperature and load. While electricity demand increases in winter due to heating requirements, hot summer days can likewise result in elevated consumption as cooling systems operate more intensively, thereby increasing the need for additional electricity supply.

Beyond these structural demand patterns, electricity markets are also influenced by broader macroeconomic conditions. Economic growth, industrial activity and business cycles affect electricity consumption through changes in production levels, investment activity and household demand, linking electricity prices to the general macroeconomic environment. Periods of economic expansion tend to raise electricity demand and tighten capacity margins, while economic slowdowns reduce load growth and ease price pressure (Edwards, 2023, p. 51 & 114).

As with demand, the supply side of the electricity market is shaped by a combination of technological, operational, meteorological and regulatory factors, which together determine both the maximum available generation and the marginal cost at which it can be supplied.

A central supply-side driver is the cost of fuels and other inputs, since these determine the marginal cost of conventional generation and thus the order in which power plants are dispatched. In economic terms, the relevant benchmark is often the levelized cost of electricity, which aggregates initial investment, fuel, operation and maintenance, and financing costs into a single unit cost over the lifetime of a plant (Mack, 2014, p. 212). For fossil-fuel technologies such as coal and gas, a large share of this cost is driven by fuel prices, so changes in coal, gas or CO_2 allowance prices feed directly into the short-run marginal cost of generation (Edwards, 2023, p. 138). By contrast, wind and solar are characterised by high upfront investment to build the plant but very low marginal costs, as additional output does not require extra fuel buying (IEA, 2025, pp. 24–25). This difference in cost structure explains why, once built, renewable plants typically enter the

merit order before conventional units and push higher-cost generators to the margin (SOURCE?).

Beyond cost considerations, the technical characteristics of generating units play an important role in determining available supply. Conventional plants differ in their ramp-up and ramp-down capabilities, minimum stable generation levels and start-up times, which constrain their ability to respond quickly to variations in demand or renewable production (Edwards, 2023, p. 28). Intermittent renewable sources such as wind and solar exhibit strong weather-related fluctuations and relatively low average availability compared with their installed capacity. For example, analyses of wind power in Germany demonstrates that only a limited portion of the installed fleet can be considered reliably available at all times, with load factors spanning a wide range between very low and moderate utilisation levels (Dannecker, 2015, pp. 18–19). Edwards (2023, p. 358) likewise notes that wind output is inherently variable and sensitive to seasonal, geographical and day time conditions. In addition to these technical and meteorological constraints, not all installed capacity is continuously usable. Parts of the generation fleet are periodically unavailable due to scheduled maintenance or unexpected outages. For this reason, adequacy assessments distinguish between unavailable and reliably available capacity, with system operators also maintaining reserve capacity to manage forecast errors and short-term imbalances (Dannecker, 2015, p. 19).

An additional factor that influences supply is the physical capacity of the transmission grid, which determines the maximum amount of electricity that can reach consumers. Even when sufficient power is produced, congestion in the high-voltage network can limit transfers between regions, requiring TSOs to intervene through redispatch measures or even grid shutdowns to maintain system stability and prevent overloads (Dannecker, 2015, p. 23). In such situations, the physical limits of the grid – or transmission congestion – lead to regional price differences, often determined by Locational Marginal Pricing (LMP), as constrained areas experience price variations depending on local supply and demand conditions. LMP is a pricing method where the price of electricity is calculated at different points across the transmission grid, instead of using one single price for different regions (Edwards, 2023, p. 133; Müsgens & Bade, 2024, p. 107). As this study focuses on price forecasting within a single market, the detailed modelling of cross-border dynamics will not be examined.

Given these structural, technical, and market-driven dynamics, the functioning of electricity markets is increasingly shaped by the regulatory frameworks that govern investment incentives, system reliability, and market behaviour. Policy instruments are essential to stabilize operations and maintain security of supply amid growing renewable penetration and heightened price volatility (IEA, 2025, pp. 7–9). Regulators have expanded the use of mechanisms such as capacity remuneration schemes to pay generators for their available capacity, ensuring sufficient dispatchable power remains available as variable renewables grow (IEA, 2025, p. 84). Simultaneously, authorities are tightening transparency and monitoring obligations – illustrated by the EU's REMIT framework – to mitigate risks linked to market manipulation and algorithmic bidding, thereby preserving market integrity in short-term markets (IEA, 2025, p. 29; Müsgens & Bade, 2024, p. 173). Current policy discussions emphasize the need for market designs and regulatory tools to align, ensuring affordability, investment certainty, and long-term system reliability (IEA, 2025, p. 15).

Beyond domestic regulation, geopolitical factors and international energy policies play an important role in shaping electricity market conditions. Decisions by major energy-producing countries and organisations, such as OPEC, influence global oil and gas prices, which indirectly affect electricity generation costs in fuel-dependent systems through price transmission mechanisms (Mack, 2014, p. 182; Edwards, 2023, p. 162). In European power systems, fossil fuel prices are particularly relevant because thermal generation technologies are often positioned at the margin of the merit order and therefore contribute directly to short-run price formation when renewable output is insufficient (Edwards, 2023, pp. 28-29). Geopolitical developments, including supply disruptions, sanctions, or changes in export policy, can thus trigger abrupt revisions in fuel price expectations and amplify price volatility in electricity markets, even though their effects materialise indirectly through input costs rather than through electricity demand itself (IEA, 2025, p. 7; Mack, 2014, p. xx).

Taken together, electricity price formation reflects the interaction of demand patterns, generation costs, network constraints, regulatory frameworks and external macroeconomic and geopolitical influences. These factors do not operate in isolation but interact in a dynamic and sometimes non-linear manner, contributing to the pronounced volatility observed in electricity markets. Recognising this complex risk environment is

essential for analysing electricity prices and provides the foundation for the forecasting approaches examined in the subsequent sections.

2.1.5 Motivation and common approaches to power trading

As demonstrated by Müsgens & Bade (2024) and Edwards electricity market instruments exhibit strong parallels to those used in traditional financial markets. While the focus of this study is on spot and day-ahead data, the market structure is also comprised of products such as forwards, futures, and other derivatives. Consequently, this section will highlight these other products to provide a comprehensive introduction, but will focus primarily on the short-term spot markets.

As established in Section XX, the day-ahead and intraday markets form the core of short-term electricity trading. The current section analyses the trading behaviors and motivations of market participants within these venues. The inherent properties of electricity – its non-storability, high volatility, and sensitivity to real-time information – are the fundamental drivers shaping the distinct and specialized trading strategies employed by market participants (Edwards, 2023, pp. 49–50; Müsgens & Bade, 2024, p. 23).

Motivations differ substantially across actor groups. Utilities and large industrial consumers primarily use short-term markets for hedging, securing expected consumption at predictable prices and thereby reducing exposure to fluctuations in spot prices and standard load profiles, which are "the average (quarter-hourly) power consumption of a typical household" (Müsgens & Bade, 2024, pp. 39–40). To complement these short-term hedges, many large consumers enter into forward contracts, which allow them to stabilise procurement costs over longer horizons by fixing part of their expected load at predetermined prices (IEA, 2024, pp. 62-63). Generation companies align their trading activity with operational optimisation. They monitor indicators such as clean spark and dark spreads, which compare electricity prices with fuel and CO_2 costs, to assess whether operating a power plant yields a positive margin. These spreads reflect the basic economics of thermal generation, where operators pay for fuel and CO_2 certificates while receiving revenue from electricity sales, making the spread a direct measure of short-term profitability. These signals inform dispatch decisions and influence whether conventional units are ramped up, reduced or temporarily shut down. Because these spreads replicate

the marginal profit of specific generation technologies, they are also traded OTC as financial products, allowing market participants to hedge or speculate on relative movements between power, fuel and emission prices (Müsgens & Bade, 2024, pp. 66–67, Edwards, 2023, p. 79).

In addition to spot and forward markets, many actors rely on exchange-traded futures and options, which offer liquidity, transparency and central clearing. Utilities and retailers predominantly use futures to manage portfolio risk in a more flexible manner than with bilateral forwards, as futures allow frequent rebalancing through daily margining and can be unwound at low transaction costs in liquid trading sessions (Mack, 2014, pp. 133–134, IEA, 2024, pp. 62-63). Generators employ futures to hedge aggregated production volumes over broader periods rather than individual delivery windows, benefiting from standardised contract sizes that simplify portfolio hedging across multiple months or seasons (Mack, 2014, pp. 141-142). Options are used when actors seek asymmetric protection: market participants buy call or put options to insure against extreme price movements, while option writers earn premiums when expecting stable market conditions (Mack, 2014, p. 143). Derivative markets therefore enable both hedgers and speculators to manage or assume risk efficiently while contributing to market liquidity (Mack, 2014, p. 167).

Other speculators in electricity markets include proprietary traders, who engage solely for financial returns without any link to physical delivery (Müsgens & Bade, 2024, p. 35), as well as power plant operators, who may pursue speculative opportunities in addition to their standard activity of producing and selling electricity. Proprietary traders and speculative desks adopt opportunity-driven strategies that respond to short-term market fluctuations. Such fluctuations often arise from unexpected forecast revisions or rapid changes in renewable production, which can create temporary price distortions in intraday spot markets (Müsgens & Bade, 2024, pp. 88–90). Their objective is to generate profit from the inherent price volatility in electricity markets, where even slight discrepancies between real-time generation and consumption can prompt price fluctuations, which in turn can be used as trading opportunities (Edwards, 2023, p. 51 & 241).

The growing operational complexity of electricity systems has also encouraged the use of algorithmic order execution within the intraday market. Automated systems monitor updated weather forecasts, TSO data and plant availability, enabling rapid adjustments to

intraday positions and helping to minimise balancing costs for portfolio managers. Such algorithms are often used to close deviations in balancing groups immediately following forecast changes. Nevertheless, discretionary trader judgement remains essential when market conditions deviate from typical patterns or when qualitative information, such as unexpected outage announcements or operational restrictions, must be interpreted (Müsgens & Bade, 2024, pp. 153-154).

Given these behavioural and technological aspects, trading outcomes are further shaped by the risk environment characteristic of electricity markets, as outlined in Section XY. High intraday volatility, frequent price spikes and uneven liquidity across products require disciplined management of mark-to-market exposures and position limits (Mack, 2014, p. 8). As Mack (2014, p. 8) notes, traders must account not only for price risk but also for operational uncertainties arising from outages and transmission congestion, as these events can rapidly alter expected market conditions. The need to react to such risks in real time places considerable emphasis on timely information processing and on the ability to adjust trading strategies swiftly in response to changing fundamentals.

Taken together, these trading practices show that electricity markets depend heavily on rapid information processing, continuous position management and flexible optimisation. They also provide a strong motivation for incorporating news-based and machine-learning approaches into price forecasting, as examined in the following sections.

2.2 Fundamentals of Natural Language Processing

2.2.1 Introduction to NLP

Natural Language Processing (NLP) is a subfield of machine learning that studies how computers can be used to analyse, interpret and generate human language in textual or spoken form (Lee, 2023, pp. 3–4). In contrast to structured numerical data, natural language is inherently unstructured, ambiguous and context dependent. The meaning of linguistic expressions depends not only on individual words but also on grammatical structure, surrounding context and domain-specific conventions, which makes automated processing particularly challenging (Lee, 2023, p. 5).

In economic and financial research, NLP has gained importance due to the increasing availability of large volumes of textual information, including news articles, policy statements and corporate disclosures, which contain information that is not immediately reflected in quantitative market data (Das, 2022, pp. 3–4). Financial markets are influenced not only by realised economic outcomes but also by expectations, uncertainty and narratives communicated through language. NLP provides a systematised framework for the conversion of unstructured textual information into quantitative representations. These representations can then be integrated into empirical models and decision-making processes, which can be employed for the generation of trading signals (Das, 2022, pp. 4–5).

Earlier approaches relied primarily on frequency-based representations, whereas modern NLP increasingly applies neural language models that can capture contextual meaning and semantic relationships more effectively (Jurafsky & Martin, 2023, p. 4). Recent progress in representation learning, particularly through transformer-based architectures, has substantially improved the ability of NLP models to account for context-dependent semantics (Jurafsky & Martin, 2023, p. 171). These models produce dense numerical representations, commonly referred to as embeddings, which encode both lexical meaning and contextual relationships within a text. Such advances have expanded the applicability of NLP to information-sensitive domains such as finance and energy markets, where small changes in wording may signal shifts in expectations or market conditions (Lee, 2023, p. 118; Das, 2022, p. 4; Simonian, 2025, p. 127)

2.2.2 Conceptual Foundations of Text-to-Quantitative Transformation

- Text Representation and Semantic Meaning

A fundamental task in Natural Language Processing is the transformation of unstructured textual data into numerical representations that can be analysed using statistical and machine-learning methods. Unlike numerical variables, text does not possess an inherent quantitative structure, which requires explicit encoding schemes that preserve linguistic meaning while enabling mathematical processing (Jurafsky & Martin, 2023, p. 4). The objective of text representation is therefore to map words, sentences or documents into vector spaces in which semantic similarity can be expressed through numerical distance (Jurafsky & Martin, 2023, p. 99).

- From Words to Vectors: Embeddings

Early approaches to text representation rely on frequency-based techniques, such as bag-of-words and term-frequency measures, which describe documents by counting word occurrences without considering word order or context (Jurafsky & Martin, 2023, p. 4 & 235). To improve their descriptive power, term weighting schemes such as Term Frequency–Inverse Document Frequency (TF-IDF) assign higher importance to words that are frequent within a document but rare across the corpus, thereby reducing the influence of common yet uninformative terms (Simonian, 2023, p. 129; Das, 2022, pp. 13 & 81). While these methods are transparent and computationally simple, they fail to capture semantic relationships between words and texts. More advanced representation methods address this limitation by embedding textual elements into continuous vector spaces based on their contextual usage, thereby allowing semantically related expressions to be represented more closely (Jurafsky & Martin, 2023, p. 95; Lee, 2023, p. 118) Embedding-based representations therefore outperform frequency-based approaches because they capture meaning through contextual co-occurrence patterns rather than isolated word counts. Words or texts that appear in similar contexts are mapped to nearby positions in the vector space, even if they do not share identical vocabulary (Jurafsky & Martin, 2023, p. 4 & 101). In practice, semantic similarity is encoded through geometric relationships, where cosine similarity or Euclidean distance between vectors provides a quantitative measure of conceptual relatedness, enabling downstream tasks such as classification, clustering and forecasting. While Euclidean distance calculates the absolute spatial magnitude between two points, cosine similarity evaluates the angular

orientation of vectors, offering a robust metric that remains independent of document length or vector magnitude (Lee, 2024, pp. 326–327).

- Transformer Models and Contextual Embeddings

While embedding-based representations capture semantic similarity through vector geometry, early embedding models assign a single representation to each word, regardless of its linguistic context. Transformer-based language models extend this framework by producing contextual embeddings, where word representations are dynamically adjusted based on surrounding tokens within a sentence (Jurafsky & Martin, 2023, p. 171). This architecture enables the model to learn complex pairwise relationships and long-range dependencies, producing contextual embeddings that adjust dynamically to a word's usage. These dynamic representations encode rich semantic and syntactic information that can be directly applied to classification and topic modelling. By addressing challenges such as polysemy, where a single word can have multiple meanings, transformers are particularly effective in financial applications. They allow for the detection of subtle contextual cues – such as shifts in tone, uncertainty, or narrative framing – that often precede observable market movements (Jurafsky & Martin, 2023, p. 172 & 206; Das, 2022, p. 4).

- Classification and Topic Modelling: Conceptual Foundations

While text representation converts language into numerical form, the analysis of this data typically follows two distinct paradigms: classification and topic modelling. Classification is a supervised learning task where documents are assigned to predefined categories based on labeled training data. Typical applications include sentiment classification, event detection or relevance filtering, where the goal is to map textual inputs to known categories (Lee, 2023, pp. 336-337; Jurafsky & Martin, 2023, pp. 61-62). In contrast, topic modelling is an unsupervised approach that identifies latent thematic structures within a corpus without prior labelling. Rather than assigning texts to explicit categories, topic models identify clusters of semantically related terms that co-occur frequently across documents, thereby revealing dominant themes present in the data (Lee, 2023, p. 125; Das, 2014, p. 42). In financial research, these techniques enable the systematic transformation of qualitative narratives – such as news flows and policy communications – into structured signals that can be linked to quantitative outcomes, making them particularly valuable for discovering emerging risks that have not yet been

captured by traditional market indices (Textual Analysis in Finance, 2020, p. 2; Simonian, 2025, p. 127).

- Dimensionality Reduction in NLP

The high dimensionality of modern text embeddings introduces computational and statistical challenges for both classification and topic modelling. Contextual embeddings generated by transformer models typically inhabit high-dimensional vector spaces, which can impede the efficiency and accuracy of clustering algorithms due to the "curse of dimensionality" (Grootendorst, 2022, p. 2). To address this, dimensionality reduction techniques are employed to project these embeddings into a lower-dimensional space while preserving their essential topological structure. The Uniform Manifold Approximation and Projection (UMAP) algorithm is frequently preferred for this task, as it effectively maintains both local and global semantic relationships, ensuring that semantically similar documents remain grouped together in the reduced space (McInnes et al., 2020, p. 1-2).

- Clustering in Text Analysis

Once embeddings are represented in a reduced space, clustering algorithms can be applied to identify groups of semantically related texts. In contrast to supervised classification, clustering aims to uncover latent structure without predefined labels, which is especially relevant when analysing large and heterogeneous document collections (Das, 2022, p. 42). Density-based approaches such as HDBSCAN are well suited for this purpose, as they can identify clusters of varying size and density while explicitly modelling noise, a common feature of real-world textual datasets (Grootendorst, 2022, p. 3; McInnes et al., 2017, pp. 1–2). Together, dimensionality reduction and clustering form a critical conceptual bridge between high-dimensional embeddings and interpretable topic structures in NLP applications.

- Topic Representation and class-based TF-IDF

The final step in the transformation process is the extraction of interpretable labels from the identified clusters. While standard TF-IDF evaluates word importance at the document level, the class-based TF-IDF (c-TF-IDF) procedure treats all documents within a cluster as a single class. This allows the model to calculate a score that reflects how representative a term is of a specific topic by comparing its frequency within the

cluster to its frequency across the entire corpus. By identifying words that are unique to specific clusters, c-TF-IDF provides a precise linguistic description of the underlying market narratives or economic themes captured by the model (Grootendorst, 2022, pp. 3–4).

- BERTTopic

The conceptual process described in this section is implemented using Bidirectional Encoder Representations from Transformers (BERT), a transformer-based language model that jointly performs tokenisation, contextual embedding generation and semantic representation (Devlin et al., 2019, pp. 1-2; Jurafsky & Martin, 2023, p. 171). BERT encodes linguistic meaning as dense vectors that capture both semantic and syntactic relationships through contextualised representations rather than fixed word-level statistics (Jurafsky & Martin, 2023, p. 197).

This research paper employs specialised variants of the BERT architecture that are tailored to the computational and analytical requirements of large-scale text analysis. Compact models such as MiniLM reduce computational complexity while preserving much of the semantic representation capacity, making them suitable for large-scale text analysis (Wang et al., 2020, p. 1). In addition, zero-shot classification models based on transformer embeddings enable semantic categorisation without task-specific labelled data, relying on general language understanding acquired during pretraining (Yin et al., 2019, pp. 1-2). The specific implementation choices and empirical integration of MiniLM and zero-shot are presented in detail in Chapter 3.

2.2.3 Limitations and Challenges of NLP in Financial Applications

Despite its growing relevance, the application of Natural Language Processing in financial contexts faces several methodological and practical limitations. A central challenge arises from the ambiguity and context sensitivity of financial language, where identical terms may carry different meanings depending on macroeconomic conditions, institutional settings or market regimes. This linguistic complexity increases the risk of misinterpretation, particularly when models are applied across heterogeneous sources such as news articles, policy statements and analyst commentary (Das, 2022, p. 105-107).

Another limitation concerns data quality and representativeness. Financial text data often contain noise, redundancy and reporting bias, as media coverage tends to concentrate on

extreme events or prominent institutions. Such distortions can lead NLP models to overemphasise sensational narratives while underrepresenting gradual or structural developments, potentially weakening the stability of derived signals (Textual Analysis in Finance, 2020, p. 3; Das, 2022 p. 105). Furthermore, changes in language use over time, driven by regulatory shifts or evolving market discourse, may reduce model performance if not adequately addressed through retraining or adaptive techniques (Loughran and McDonald, 2020, pp. 4-5).

From a modelling perspective, transformer-based architectures introduce challenges related to interpretability and computational cost. While contextual embeddings offer superior semantic representation, their internal decision processes remain largely opaque, complicating economic interpretation and model validation (Wang et al., 2020, p. 1; Das, 2022, p. 107; Jurafsky & Martin, 2023, p. 172). In addition, the reliance on large pretrained models raises concerns regarding domain mismatch, as models trained on general language corpora may not fully capture specialised financial or energy-market terminology without further adaptation (Kaplan et al., 2023, p. 1; Das, 2022, p. 105; Jurafsky & Martin, 2023, p. 162).

Taken together, these limitations highlight the importance of careful model selection, robust preprocessing and complementary quantitative validation when deploying NLP techniques in financial and energy market research.

2.3 Fundamentals of Trade Signal Generation using NLP

Building upon the linguistic foundations and transformer-based representations introduced in Section 2.2, this chapter examines the practical application of Natural Language Processing within financial and energy markets. The literature reviewed in this section reflects a clear methodological distinction between forecast generation and trading implementation.

Section 2.3.1 focuses on studies that integrate textual information into forecasting frameworks, where NLP-derived features are used to improve predictions of prices, returns, volatility, or spreads. These contributions primarily evaluate performance using statistical accuracy metrics and demonstrate that textual data enhances predictive power

beyond traditional quantitative inputs. However, most of this literature does not translate forecasts into explicit trading decisions.

Section 2.3.2 therefore addresses a subsequent and conceptually distinct step: the transformation of NLP-based forecasts into actionable trading signals. This section reviews approaches that incorporate confidence measures, threshold rules, and risk controls to convert model outputs into long, short, or hold positions. Particular attention is given to empirical applications in energy and electricity markets, where rapid information processing and expectation-driven price dynamics make NLP-based trading strategies especially relevant.

2.3.1 From Textual Information to Market Forecasts

Building on the power market characteristics discussed in Section 2.1 and the text-to-vector transformation outlined in Section 2.2, recent research integrates textual information into forecasting models for energy and commodity prices. Standard forecasting setups typically rely on historical prices, load patterns, weather variables, and physical system constraints, which capture structural and seasonal behaviour but only partially reflect forward-looking information (Hong et al., 2020, pp. 377–378; Müsgens & Bade, 2024, pp. 5–6). Textual sources such as news and policy communication can contain expectation- and uncertainty-related signals that precede observable price adjustments, making them a natural complement to purely quantitative inputs (Das, 2014, p. 4; Loughran and McDonald, 2020, pp. 1-2). In energy markets, this channel is particularly relevant because regulatory and geopolitical developments may shift price expectations before physical effects materialise (Dannecker, year, pp. 2-3; Edwards, 2023, p. 162).

Early empirical studies linking news to market outcomes focused on sentiment indicators derived from textual data. Lee Smales (2014) examines the gold futures market using commodity-specific news sentiment scores obtained from structured news analytics. The study aggregates sentiment indicators extracted from financial media and incorporates them into regression-based forecasting models alongside historical price information news (Smales, 2014, pp. 277–278). Using daily gold futures data over an extended sample period, Smales (2014, pp. 285-286) shows that news sentiment significantly improves the prediction of short-term returns and volatility, with negative news exerting a stronger market impact than positive news.

Similarly, Li et al. (2021) analyse oil futures markets and construct sentiment measures from oil-related news articles. Their framework combines textual sentiment indicators with traditional econometric volatility models, demonstrating that sentiment contains incremental predictive information for both returns and volatility, particularly during periods of elevated uncertainty (Li et al., 2021, pp. 3–4). These early contributions treat sentiment as a low-dimensional numerical signal derived from text, consistent with the frequency-based and term-weighting approaches discussed in Section 2.2.

Subsequent research extends this idea by embedding textual features directly into forecasting models rather than relying solely on aggregated sentiment scores. Guan et al. (2022) propose a forecasting framework for international natural gas prices that combines historical price series with numerical representations of related news articles. Textual information is first transformed into quantitative features and then integrated into machine-learning models alongside lagged price inputs (Guan et al., 2022, pp. 5–6). Their empirical results show that news-augmented models outperform price-only benchmarks, especially in capturing abrupt price movements associated with geopolitical events and supply disruptions (Guan et al., 2022, p. 10). This approach illustrates how text embeddings, as introduced in Section 2.2, can be treated as additional explanatory variables within standard forecasting pipelines.

More recent studies leverage advances in transformer-based language models to extract richer semantic representations from news. Chakraborty et al. (2024, pp. 3-4) develop an event-driven forecasting framework in which news articles are encoded using contextual embeddings derived from transformer models. Their Robust Event Network (REN) model is a neural network-based predictive model that specifically utilizes Long Short-Term Memory (LSTM) hidden states to capture non-linear, long-term dependencies between extracted event intensities and commodity price fluctuations (Chakraborty et al., 2024, pp. 6 & 13). Rather than modelling sentiment alone, this framework explicitly links semantic content to price reactions. This aligns with the argument of Kaplan et al. (2023) that general language models require asset-specific adaptation to capture economically relevant information in energy and commodity markets (Kaplan et al., 2023, p. 1). The authors demonstrate that incorporating exogenous event features from news significantly improves accuracy for commodities sensitive to real-world developments. They highlight that while traditional models often fail to account for extreme volatility, their approach captures the impact of external shocks and provides early warning signals for abrupt price

fluctuations following significant information releases. (Chakraborty et al., 2024, pp. 24-25).

While RENs rely on deep learning architectures, other high-performing frameworks in financial forecasting frequently employ gradient-boosted decision trees to integrate these textual embeddings with numerical data. Methods such as XGBoost are particularly effective in this context as they iteratively combine multiple weak learners into a strong predictive model through additive training, minimizing forecast errors while efficiently capturing complex, non-linear interactions (Chen & Guestrin, 2016, pp. 1-2; Simonian, 2025, pp. 54 & 65).

Within electricity markets, the relevance of NLP-based forecasting has been demonstrated more explicitly. Bai et al. (2024, pp. 1-2) investigate the joint role of news and load information in electricity price forecasting. Their framework combines traditional physical variables, such as demand and load forecasts, with textual features extracted from energy-related news. Using supervised learning models, they show that incorporating news-derived signals improves forecast accuracy relative to models relying solely on load, weather and historical prices. The results underline that news captures regulatory, infrastructural and geopolitical information that is not directly observable in physical system data, reinforcing the complementarity between NLP-based features and established electricity market fundamentals (Bai et al., 2024, pp. 5–6).

Bi et al. (2025, pp. 1-2) further advance this line of research by introducing a large-scale, news-augmented benchmark dataset for electricity price forecasting using large language models. News texts are encoded using transformer-based embeddings and integrated into forecasting architectures across multiple time horizons (Bi et al., 2025, pp. 6). Their empirical analysis demonstrates that NLP-augmented models deliver improvements in predictive performance, particularly during periods of market stress and heightened volatility (Bi et al., 2025, pp. 7 & 9). This study provides systematic evidence that textual representations derived from large language models are not merely auxiliary inputs but can become central components of modern electricity forecasting frameworks (Bi et al., 2025, pp. 2-3).

Taken together, the literature demonstrates a clear evolution from simple sentiment indicators towards embedding-based and event-driven forecasting frameworks. Textual information is increasingly treated as a quantitative input that complements historical

prices and physical fundamentals. In energy markets in particular, where prices react strongly to expectations and policy-driven events, NLP-based forecasting models provide a natural extension to established time-series approaches discussed by Dannecker (2015) and Müsgens and Bade (2024). This body of research forms the conceptual foundation for the signal-generation framework developed in the subsequent Chapter 3.

Maybe add a paragraph about the limitations?

2.3.2 From Forecasts to Trading Signals

While the literature reviewed in Section 2.3.1 focuses on improving predictive accuracy, forecasting performance alone does not automatically translate into economically meaningful trading outcomes. The transformation of forecasts into trading signals requires an additional decision layer that accounts for market frictions, risk constraints, and asymmetric payoff structures. In practice, market participants must determine when a forecast is sufficiently strong to justify a position and how it should be implemented within the specific microstructure of the traded market (Mack, 2014, pp. 7–9; Müsgens & Bade, 2024, pp. 22–23).

A common approach in the literature is to convert point forecasts or directional predictions into discrete trading actions, such as long, short, or neutral positions. This step typically involves the use of confidence thresholds or probability cut-offs, whereby positions are only entered when the model's signal exceeds a predefined level of certainty. Such threshold-based decision rules help mitigate overtrading and reduce exposure to noise-driven forecasts, a consideration that is particularly important in volatile and illiquid segments of electricity markets (Müsgens & Bade, 2024, pp. 33 & 44; Chakraborty et al., p. 7; Edwards, 2023, p. 51 & 270).

Recent studies applying NLP-based models increasingly adopt this two-stage framework. Kirtac and Germano (2024) explicitly separate forecast generation from signal execution by first producing sentiment-based directional predictions using transformer embeddings and subsequently applying confidence filters to derive trading positions. Their empirical results show that profitability is highly sensitive to signal calibration: strategies with stricter confidence thresholds exhibit lower trade frequency but achieve superior risk-adjusted performance, as measured by Sharpe ratios and drawdown metrics (Kirtac & Germano, 2024, pp. 9–10). Kemal similarly demonstrates that large language model outputs can be transformed into economically viable trading strategies when combined

with disciplined position sizing, volatility targeting, and drawdown constraints. Rather than trading on every forecast, Kemal shows that selective execution materially improves performance stability across market regimes (Kemal, year, pp. xx–xx).

In energy markets, the conversion from forecasts to trades must also reflect physical and institutional constraints. Electricity prices are subject to sharp intraday reversals, negative pricing, and uneven liquidity across delivery periods (Section XY). As noted by Dannecker (2015, pp. 20–21), trading strategies that ignore these features risk systematic losses despite statistically accurate forecasts. Consequently, practical implementations often restrict trading to specific time windows, delivery products, or volatility regimes, aligning model outputs with operational feasibility.

Interpretability plays a particularly important role in this context. Electricity traders frequently rely on discretionary judgement when reacting to unexpected events such as outages, regulatory announcements, or geopolitical developments. Müsgens and Bade (2024, pp. 88–90) emphasise that qualitative understanding of why a price signal arises is often as important as the signal itself. Topic-based representations derived from c-TF-IDF, as introduced in Section 2.2.2, support this requirement by providing transparent keyword descriptions for each identified topic. This allows traders to distinguish, for example, between signals driven by "supply outages" and those linked to "macroeconomic conditions" (Grootendorst, 2022, pp. 3–4). Such interpretability facilitates discretionary overlays and reduces the risk of mechanically executing opaque model outputs.

From a methodological perspective, machine-learning models serve as an intermediate layer between NLP-derived features and final trading decisions. In line with the forecasting frameworks discussed in Section 2.3.1, gradient-boosted decision tree models such as XGBoost are frequently employed to map high-dimensional textual features into probabilistic forecasts or directional indicators (Chakraborty et al., 2024, pp. 7–8; CFA Institute, 2023, p. 56). In trading applications, these model outputs are not executed directly but are typically embedded within rule-based decision logic, where confidence thresholds, position sizing rules, and market-specific constraints determine whether and how a forecast is translated into a tradable position.

A critical practical limitation in electricity markets concerns signal timeliness. Price adjustments can occur within minutes following information releases, particularly in

intraday markets. Transformer-based models, while semantically powerful, entail substantial computational costs that may introduce latency between news arrival and signal generation. This creates a trade-off between semantic richness and signal velocity, where more complex models risk producing signals that decay before execution becomes economically viable (Das, 2022, p. 107; Jurafsky & Martin, 2023, p. 172). As a result, model design must balance representational depth against computational efficiency, especially in short-horizon trading contexts.

Taken together, the literature indicates that the economic value of NLP-based forecasts depends critically on the design of the signal-generation layer. Effective strategies do not trade on predictions mechanically but embed them within disciplined execution frameworks that account for confidence, interpretability, market structure, and latency constraints. This distinction between forecasting accuracy and tradable signals motivates the two-stage modelling approach adopted in this study, where NLP-derived forecasts are subsequently transformed into actionable long, short, or hold decisions under explicit decision rules, as detailed in Chapter 3.

3 Methodology

3.1 Framework Structure

- Introduction to our Data & Methodology pipeline
- Showcase an overview of the process used (flowchart) from data sourcing all the way to model evaluation & backtesting.
- Explain and highlight which models will be used (XGBoost, Ridge & Logistic Regression). We use a dual model set-up, where we first classify the next day spread to be positive or negative alongside a confidence score, then use a Logistic (or lightGBM) to then turn those forecast into long/short/hold decisions
- Quickly explain how the infrastructure was set up (GPU clustering, code optimisation for parallel processing)
- Give our model a name (ex. PowerTree)

3.2 Data Sourcing

This research project integrates two complementary data streams: textual news data capturing market sentiment and discourse, and quantitative electricity market data

reflecting trading dynamics. The sourcing strategy prioritised comprehensive coverage across both domains whilst maintaining data quality and the temporal alignment necessary for signal generation. Electricity markets operate at the intersection of physical commodity trading and information-driven financial markets, where news events can trigger immediate price responses whilst fundamental supply-demand dynamics establish longer-term trends (Ziel & Steinert, 2024). The following subsections detail the collection procedures for news data (Section 3.2.1) and energy market data (Section 3.2.2), before Section 3.3 describes the preprocessing pipeline that integrates these sources.

3.2.1 News Data Collection

News data were collected using NewsAPI, an aggregation service providing programmatic access to historical and current news articles from diverse sources. NewsAPI was selected over alternative services (such as GDELT or proprietary Bloomberg feeds) due to its comprehensive metadata structure, including timestamps, source attribution, and article descriptions, which enables precise temporal alignment with market data. The service's structured data format has been employed in recent commodity market forecasting applications to extract market-relevant signals from high-frequency news flows (SOURCE).

The news collection strategy encompassed both English- and German-language sources to capture the full spectrum of discourse surrounding German electricity markets. German-language sources provide domestic market insights, regulatory announcements, and regional policy developments, whilst English-language international sources offer broader market analysis, global energy trends, and institutional investor perspectives that influence local prices (SOURCE). This bilingual approach is particularly relevant given Germany's central role in European electricity markets, where cross-border flows and interconnector capacity create price dependencies across multiple national markets (SOURCE). All collected articles were subsequently translated into German (as detailed in Section 3.3) to enable consistent downstream natural language processing.

The collection period spanned from 1 November 2020 to 31 October 2025, providing a five-year window that captures significant market regime changes: the COVID-19 pandemic's demand disruptions, the 2022–2023 energy crisis triggered by geopolitical tensions following Russia's invasion of Ukraine, and the subsequent market stabilisation accompanied by accelerated renewable deployment (SOURCE). This temporal coverage

exposes the model to diverse market conditions, including both low-volatility and crisis periods, thereby improving robustness to the regime changes that characterise modern electricity markets.

The retrieval strategy employed daily queries using NewsAPI's premium subscription tier, which permits a maximum of 100 articles per API call. Each day's query retrieved articles matching the specified keywords and source parameters (as detailed in Table XX). On days when more than 100 matching articles were published, the API returned the 100 most relevant results ranked by NewsAPI's relevance algorithm; however, such instances were rare across the five years. This systematic daily collection methodology aligns with best practices for constructing news-based trading signals, where consistent temporal coverage prevents information gaps that could compromise signal quality (Smales, 2014, pp. 277–279). The collection contained 89,517 English-language and 122,208 German-language articles before preprocessing.

Table 1: News Search Parameters

Parameter	*English Sources*	*German Sources*
News Outlets	ABC News, Associated Press, BBC News, Bloomberg, CNN, MSNBC, NBC News, Reuters, The Wall Street Journal, The Washington Post, Time, USA Today	Bild, Der Tagesspiegel, Die Zeit, Focus, Handelsblatt, Spiegel Online, WirtschaftsWoche
Keywords	EPEX, Erdgas, Erdöl, EU, EZB, IEA, Kohle, OPEC, Strombedarf, Stromeffizienz, Stromerzeugung, Stromnachfrage, Stromsparen, Wetter, Zentralbank	ACER, Carbon, Climate, Coal, ECB, Electricity, Emissions, Energy, EPEX, European Comission, European Union, Gas, Grid, Hydro, IEA, Infrastructure, Interest Rates, Market, Nuclear, OECb, OPEC, Oil, Pipeline, Policy, Renewable, Solar, Storage, Supply, Tariffs, Transmission, Wind
Articles Retrieved	89’517	122’208
Timeframe	1 November 2020 - 31 October 2025	1 November 2020 - 31 October 2025

The keyword selection strategy (detailed in Table XX) balanced domain specificity with comprehensive coverage, incorporating terminology from energy infrastructure, market mechanisms, relevant institution abbreviations, and macroeconomic indicators. This multi-faceted approach recognises that electricity prices respond to diverse information sources, ranging from technical factors such as renewable generation forecasts to broader economic indicators that affect industrial demand (QUELLE AUS TEIL OBEN).

3.2.2 Energy Market Data Collection

Energy market data were sourced from Energy-Charts.info, a platform maintained by the Fraunhofer Institute for Solar Energy Systems (ISE) that has become a widely used resource in German electricity market research (Wirth, 2023). The platform aggregates data from multiple authoritative sources, including ENTSO-E, EPEX SPOT, and Germany's four transmission system operators, providing harmonised datasets that would otherwise require complex reconciliation across different source systems (Fraunhofer ISE, 2024; Schmidt et al., 2023).

Data were retrieved at 15-minute granularity for the period 1 November 2020 to 31 October 2025, matching the news collection timeframe. The following price variables were collected: day-ahead auction prices (EUR/MWh) representing the uniform clearing prices established in the daily EPEX SPOT auction, and intraday continuous market prices (EUR/MWh) reflecting real-time trading activity. These two price series form the basis for computing the spot-day-ahead spread that serves as the target variable for this study.

Generation volumes were retrieved, disaggregated by source category to capture the supply-side dynamics that influence price formation. The dataset includes output from onshore wind, offshore wind, solar photovoltaic, nuclear, lignite, hard coal, natural gas, pumped-hydro, and run-of-river hydro facilities (SOURCE). This granular breakdown enables the model to account for the merit-order effects discussed in Section 2.1, where low-marginal-cost renewables displace higher-cost thermal generation and compress electricity prices during periods of high renewable feed-in. Total electricity demand (MW) was also collected, representing the aggregate load that must be balanced against generation in real time.

The raw dataset comprised approximately 175,000 observations at 15-minute resolution. These data were converted from the original JSON format to CSV for subsequent preprocessing. Quality assessment of the raw data identified several categories of missing or anomalous values. Gaps in price reporting occurred during approximately 0.4% of 15-minute intervals, concentrated around daylight saving time transitions, exchange system maintenance windows, and periods of extreme market stress when trading was temporarily suspended. These data quality issues were addressed during the preprocessing stage described in Section 3.3.

3.3 Data Preprocessing

The preprocessing pipeline comprised two parallel workflows: one for news data and one for energy market data, which were subsequently integrated to construct the final analytical dataset. This section details the procedures employed to ensure data quality, topical relevance, and temporal alignment across these heterogeneous sources.

3.3.1 News Corpus Refinement

The broad keyword-based retrieval strategy employed in Section 3.2.1, whilst ensuring comprehensive coverage of potentially relevant content, inevitably captured substantial quantities of off-topic material. Articles matching terms such as "energy" or "market" frequently addressed topics unrelated to electricity trading, including sports events, entertainment news, and general political coverage. Rigorous filtering was therefore required to isolate energy-relevant news content suitable for signal generation.

Off-topic content was identified and removed through an iterative refinement procedure using BERTopic (Grootendorst, 2022), an unsupervised topic modelling framework combining transformer-based embeddings with hierarchical density-based clustering. BERTopic was selected for this task because it enables the discovery of latent thematic structures without requiring predefined category labels, making it well-suited to exploratory corpus analysis in which the full range of off-topic content is not known in advance (SOURCE). The framework's use of HDBSCAN clustering also accommodates topics of varying sizes and explicitly identifies noise documents that do not belong to any coherent theme.

Each iteration of the refinement procedure proceeded as follows: (1) a topic model was fitted to the corpus to discover latent thematic clusters; (2) human reviewers inspected topic labels, representative keywords, and sample documents to identify non-energy-related themes; (3) characteristic keywords and phrases associated with off-topic clusters were extracted to construct exclusion filters; (4) articles containing these keywords in titles or descriptions were removed from the corpus; and (5) the filtered corpus was resubmitted for subsequent refinement. This human-in-the-loop approach ensured that automated clustering was validated against domain expertise before content removal, preventing the erroneous exclusion of legitimately relevant articles that might share surface-level vocabulary with off-topic content.

Nine iterative filtering rounds were applied to the English corpus, progressively removing articles associated with sports coverage (team names, match results, athlete profiles), entertainment news (film releases, celebrity coverage, television programming), crime reporting (arrests, trials, investigations), retail and consumer news (product launches, shopping trends), health and medical coverage unrelated to energy policy, weather reporting without market implications, and general political coverage lacking energy or economic relevance. This process reduced the English corpus from 89,517 to 39,744 articles, a 55.6% reduction. The substantial reduction reflects the broad semantic scope of English-language international news sources, where energy-related keywords frequently co-occur with unrelated content.

The German corpus required only three iterative rounds, targeting crime, education, sports, and tourism content, and was reduced from 122,208 to 113,894 articles, which represents a more modest 6.8% reduction. The disparity in reduction rates suggests that German-language sources retrieved via energy-specific keywords were more topical than their English-language counterparts. This likely reflects the more specialised focus of German business publications such as Handelsblatt and WirtschaftsWoche, which concentrate on economic and market coverage rather than general news. A detailed log of excluded topic categories and representative keywords per iteration is provided in Appendix X.

Following topic-based filtering, English articles were translated into German using the Google Translate API via the deep-translator Python library. Translation into German (rather than translating German content into English) was chosen for two reasons. First, German-language transformer models fine-tuned on energy and economic terminology prevalent in the DACH-region discourse were available for downstream classification tasks. Second, the majority of the filtered corpus (74%) was already in German, minimising the proportion of content subject to potential translation errors. The translated English corpus (39,744 articles) was merged with the filtered German corpus (113,894 articles), yielding a total of 153,638 articles. After removing translation failures and malformed entries, the final corpus comprised 149,512 German-language articles spanning the five-year collection period.

3.3.2 Energy Market Data Processing

Energy market data underwent preprocessing to ensure temporal alignment with the news corpus and to prepare variables for feature engineering. The raw data, retrieved at 15-minute granularity, were resampled to hourly frequency by computing means for price variables and sums for volume variables. Hourly aggregation was selected to align with the day-ahead auction structure, where prices are determined for each hour of the following day, and to reduce computational overhead whilst preserving meaningful intraday variation. Following resampling, the energy dataset comprised 43,824 hourly observations spanning the five-year collection period.

Missing values were identified in 192 hourly records, primarily during daylight saving time transitions and periods of data reporting gaps. These observations were excluded rather than imputed to preserve data integrity, as interpolation of price data during market discontinuities could introduce spurious patterns. Anomalous price observations, including negative prices below -500 EUR/MWh and favourable prices exceeding 1,000 EUR/MWh, were retained in the dataset, as these extreme values represent genuine market conditions during periods of renewable oversupply or acute scarcity rather than data errors. After excluding outliers, the processed energy dataset comprised 43,632 hourly observations.

3.4 Feature Engineering

The feature engineering process transforms raw energy market data and pre-processed news articles into a structured feature matrix suitable for predictive modelling. This section details the construction of baseline features derived from electricity market fundamentals, the extraction of topical signals via zero-shot classification, the generation of semantic embeddings from news text, and the temporal aggregation of these news-derived features using exponential-decay weighting. The resulting feature set integrates both quantitative market dynamics and qualitative information encoded in news discourse, thereby enabling the model to capture the complex interplay between market fundamentals and information flows that characterise short-term electricity price movements.

3.4.1 Baseline Features

The baseline feature set comprises variables derived directly from the energy market data described in Section 3.2.2. These features capture the fundamental supply-demand dynamics that drive electricity price formation, as well as temporal patterns that reflect the cyclical nature of electricity consumption (SOURCE 2.1).

Price and Spread Features: The primary price variables are the day-ahead auction price (EUR/MWh) and the intraday spot price (EUR/MWh), both resampled at hourly frequency. The absolute spread between these prices, computed as the difference between spot and day-ahead values, serves as the basis for the target variable. This spread reflects the deviation between anticipated and realised market conditions, capturing the information advantage that accrues between the day-ahead auction close and real-time delivery (SOURCE).

Generation and Load Features: Aggregate electricity generation and total system load (MW) provide direct measures of supply-demand balance. As discussed in Section 2.1.4, the merit-order effect implies that price formation depends critically on the marginal generating unit dispatched to meet demand. Aggregate load serves as a proxy for demand pressure, where higher consumption during peak periods typically elevates prices by requiring the dispatch of higher-cost thermal generation (Edwards, 2023, pp. 134–138).

Lagged Features: To capture autocorrelation in electricity prices and demand patterns, lagged values of key variables are included at two horizons: 24 hours and 168 hours. Specifically, the feature set contains *price_lag_24*, *price_lag_168*, *load_lag_24*, and *load_lag_168*. These lags reflect the strong daily and weekly seasonality documented in electricity market research, in which demand patterns exhibit pronounced differences between weekdays and weekends and between daytime peak hours and overnight off-peak periods (Weron, 2014, pp. 1049–1051).

Temporal Features: Categorical and cyclical time indicators encode the position of each observation within relevant periodic structures:

- *hour* (0–23): Captures intraday variation aligned with consumption patterns
- *day_of_week* (0–6): Distinguishes weekday from weekend dynamics

- *day_of_year* (1–366): Encodes annual seasonal effects, including heating and cooling demand
- *week_of_year* (1–52): Provides coarser seasonal resolution
- *month* (1–12): Captures broader seasonal trends in renewable generation and demand

Together, these 13 baseline features establish a foundation that captures the structural characteristics of electricity markets before the incorporation of news-derived signals.

3.4.2 Topic Classification via Zero-Shot Learning

The transformation of news articles into structured topical features is achieved via a zero-shot classification pipeline that assigns each article to predefined energy-relevant categories derived from section 2.1. This approach leverages the semantic generalisation capabilities of large language models, enabling classification across diverse topic labels based solely on natural language descriptions (Yin et al., 2019, pp. 1–3).

The classification pipeline employs the mDeBERTa-v3-base-xnli-multilingual-nli-2mil7 model developed by Laurer et al. (2024), which extends Microsoft's DeBERTa-v3 architecture (He et al., 2023) to multilingual zero-shot classification. DeBERTa-v3 improves upon earlier transformer architectures by employing disentangled attention mechanisms that separately encode content and positional information, combined with ELECTRA-style replaced-token detection during pre-training, achieving state-of-the-art performance on natural language understanding benchmarks (He et al., 2023, pp. 6–9).

Zero-shot classification through NLI operates by reformulating the classification task as textual entailment(Yin et al., 2019, pp. 1–4). For each candidate label, the model evaluates whether a hypothesis constructed from the label is entailed by the input text (premise). The hypothesis template employed is:

"Der Artikel handelt von: {label}." (English: "The article is about: {label}.")

The model outputs entailment probabilities for each label, and the label with the highest probability is assigned as the classification. This formulation enables flexible classification across arbitrary label sets without retraining, a property particularly valuable when domain-specific labelled data are scarce or costly to obtain (Laurer et al., 2024, pp. 3–6).

The topic taxonomy was designed to capture the primary drivers of electricity price variation identified in Section 2.1 and was organised into a two-stage hierarchical structure that balances classification precision with computational efficiency. The first stage routes articles to one of seven high-level categories, while the second stage assigns articles to specific leaf topics within the selected category.

The seven high-level categories reflect distinct channels through which news content may influence electricity prices:

- *Nachfrage* (Demand): News affecting electricity consumption levels
- *Angebot* (Supply): News concerning generation capacity and availability
- *Brennstoffpreise* (Fuel Prices): News on input costs for thermal generation
- *Makrofinanzen* (Macro Finance): Broader economic conditions affecting markets
- *Geopolitik* (Geopolitics): International developments impacting energy supply
- *Wetter* (Weather): Meteorological conditions influencing demand and renewable output
- *Sonstiges* (Other): Articles lacking clear energy market relevance

This categorisation draws on the supply and demand factors discussed in Section 2.1, where fuel prices, weather conditions, and geopolitical developments were identified as key determinants of electricity price dynamics. Within each high-level category, more granular leaf topics capture directional implications for electricity prices. The complete taxonomy can be seen in Table XX.

Category	*Leaf Topics*
Nachfrage	Electricity consumption in Germany rising; Electricity consumption in Germany falling
Angebot	Wind and solar generation rising; Wind and solar generation falling; Grid or plant outages reducing supply; Expansion of LNG terminals, pipelines, or plants increasing supply
Brennstoffpreise	Wholesale gas prices rising; Wholesale gas prices falling
Makrofinanzen	Rising interest rates or high inflation tightening markets; Falling interest rates or easing inflation calming markets
Geopolitik	Geopolitical tensions or sanctions exacerbating energy supply; Geopolitical détente or lifted sanctions reducing supply risks
Wetter	Cold, calm, or overcast conditions increasing price pressure; Mild weather, strong wind, or sunshine easing prices
Sonstiges	No relevance to energy, weather, or financial markets

Table XX:

The directional framing of leaf topics (rising/falling, increasing/decreasing) enables the model to capture not merely topical relevance but also the implied price impact direction, which is critical for generating actionable trading signals. It should be noted that certain

factors relevant to electricity price formation, such as carbon prices under the EU Emissions Trading System (EU ETS) and cross-border interconnector flows, are not explicitly represented in this taxonomy. These omissions reflect the focus on factors most directly captured in German-language news discourse and represent a limitation of the current approach.

The hierarchical classification proceeds through two inference stages with confidence-based fallback logic to handle ambiguous cases. Each article's title is processed against the seven category labels using the zero-shot pipeline. The model returns probability scores reflecting the confidence of each category assignment. If the highest-scoring category exceeds the Stage 1 threshold (τ_1 = 0.35), the article proceeds to Stage 2 classification within that category; otherwise, it is routed to the fallback category ("Sonstiges").

Within the assigned category, the article is classified against the corresponding leaf topics. If the selected category contains multiple leaf topics, a second inference pass determines the specific topic. A Stage 2 threshold (τ_2 = 0.25) governs whether the leaf assignment is retained or whether the article falls back to the default "other" topic.

Articles initially classified as "Sonstiges" (no energy relevance) undergo a secondary classification using the article description rather than the title. This fallback addresses cases in which headlines are ambiguous, but article summaries provide clearer topical signals. The reclassified results replace the initial "other" assignment where a more specific topic achieves sufficient confidence.

The threshold values (τ_1 = 0.35, τ_2 = 0.25) were determined through manual inspection of classification score distributions, balancing the trade-off between coverage (retaining more articles in specific categories) and precision (avoiding misclassification of ambiguous content), following the calibration approach discussed by Laurer et al. (2024). Lower Stage 2 thresholds reflect the observation that within-category classification tends to produce lower absolute confidence scores due to the semantic similarity among leaf topics within the same category.

Articles classified as having no energy relevance are excluded from subsequent time-decay aggregation, ensuring that the news-derived features reflect only content with plausible market relevance. This filtering step is essential, given the broad keyword-based

data-sourcing strategy employed, which inevitably captured some off-topic content despite the preprocessing refinements described in Section 3.3.1.

3.4.3 Sentence Embedding Generation

Complementing the discrete topical classification, dense sentence embeddings capture semantic information that predefined topic categories may not fully represent. While zero-shot classification assigns articles to discrete labels, embeddings encode the full semantic content of article text as continuous vectors, preserving nuanced information that could inform price predictions.

The embedding pipeline employs the *paraphrase-multilingual-MiniLM-L12-v2* model from the Sentence Transformers library (Reimers & Gurevych, 2019, pp. 1–2). This model generates 384-dimensional dense vector representations optimised for semantic similarity tasks across 50+ languages, including German.

The architecture builds upon the MiniLM knowledge distillation framework (Wang et al., 2020, pp. 2–5), which compresses larger transformer models while preserving most of their representational capacity. MiniLM achieves this compression by training a smaller student model to mimic the self-attention distributions of a larger teacher model, yielding compact models suitable for large-scale text processing without substantial performance degradation.

This multilingual capability extends monolingual sentence embeddings to multiple languages through knowledge distillation on parallel corpora (Reimers & Gurevych, 2020, pp. 1–3). A fixed English teacher model produces target embeddings, while the student model learns to map both English text and its translations to the exact vector representation. This approach ensures that semantically equivalent content in different languages maps to similar positions in the embedding space.

For each news article, the embedding is computed from the article title using mean pooling over the contextualised token representations produced by the final transformer layer. Titles were selected over full article text for computational efficiency, as headlines typically convey the primary informational content relevant to rapid market reactions. However, this approach may lose nuanced information contained in article bodies, representing a trade-off between computational tractability and information completeness.

The output of this stage is a 384-dimensional embedding vector for each article, stored alongside the article metadata for subsequent temporal aggregation.

3.4.4 Time-Decay Aggregation of News Features

Individual article-level classifications and embeddings must be aggregated to the hourly resolution of the energy market data to enable joint modelling. The aggregation procedure employs exponential time-decay weighting, which assigns greater influence on more recent articles while retaining diminishing contributions from older content within a lookback window. This approach reflects the intuition that news impact on market expectations decays over time as information is absorbed and superseded by subsequent developments.

The application of exponential decay weighting to news signals draws on established practices in financial forecasting and time-series analysis. Studies demonstrated that media content influences investor sentiment and stock returns, with the effect diminishing over subsequent trading days as information is incorporated into prices(Tetlock, 2007, pp. 1149–1151). Further work in commodity markets demonstrated that incorporating time-decayed news sentiment significantly improves crude oil price forecasting, with the decay parameter governing the model's effective memory horizon (Li et al., 2021, pp. 3–4).

In electricity markets, the high frequency of price dynamics and the rapid incorporation of information suggest that news relevance diminishes relatively quickly. Forecast revisions for renewable generation, updates to plant availability, and shifts in demand expectations are continuously reflected in intraday trading, implying that older news articles provide progressively less marginal information (Weron, 2014, pp. 1031–1032).

The exponential decay formulation provides a parsimonious representation of this information decay:

$$w_i = e^{-\lambda \cdot h_i}$$

Where w_i is the weight assigned to the article i, λ is the decay rate parameter, and h_i is the number of hours elapsed since the article i was published. Higher values of λ produce faster decay, concentrating influence on very recent articles, while lower values extend the effective memory horizon, allowing older articles to retain meaningful weight.

For each hourly timestamp in the energy dataset, time-decayed weighted counts are computed for each of the 14 leaf topics. The aggregation proceeds as follows:

1. **Define Lookback Window**: For timestamp t, identify all articles published within the interval $(\mathrm{t} - \mathrm{L}, \mathrm{t}]$, where L is the lookback window in hours.
2. **Compute Decay Weights**: For each article i in the window, calculate the decay weight $\mathrm{w_i} = \mathrm{e}^{-\lambda \cdot \mathrm{h_i}}$, where $\mathrm{h_i} = \mathrm{t} - \mathrm{t_i}$ is the number of hours since the time of publication.
3. **Aggregate by Topic**: For each topic k, compute the weighted count as $\mathrm{topic_count_k(t)} = \sum_{\mathrm{i} \in \mathrm{topic\ k}} \mathrm{w_i}$
4. **Filter Irrelevant Articles**: Articles classified as "Sonstiges" are excluded from the aggregation, ensuring that topic features reflect only energy-relevant news content.

The output is a matrix of dimension $(\mathrm{T} \times 14)$, where T denotes the number of hourly timestamps, and each column represents the time-decayed count for a given topic. This aggregation preserves the interpretability of topical features: higher values indicate greater recent news volume on that topic, with recency weighted. The directional framing of topics (e.g., "gas prices rising" vs. "gas prices falling") enables the model to distinguish between news flows with opposing price implications.

Time-decayed aggregation of embeddings follows a similar procedure, with the key difference that vector averages replace scalar sums:

1. **Define Lookback Window**: Identify articles within $(\mathrm{t} - \mathrm{L}, \mathrm{t}]$ for each timestamp t.
2. **Compute Decay Weights**: Calculate $\mathrm{w_i} = \mathrm{e}^{-\lambda \cdot \mathrm{h_i}}$ for each article.
3. **Weighted Average Embedding**: Compute the aggregated embedding as $\mathrm{e_{agg}(t)} = \frac{\sum_\mathrm{i} \mathrm{w_i} \cdot \mathrm{e_i}}{\sum_\mathrm{i} \mathrm{w_i}}$, where $\mathrm{e_i}$ is the 384-dimensional embedding vector for the article i.

The weighted average ensures that the aggregated embedding lies within the convex hull of individual article embeddings, preserving semantic interpretability. If no articles fall within the lookback window for a given timestamp, the aggregated embedding is set to zero, indicating the absence of relevant news content. The output is a matrix of dimension

$(T \times 384)$, representing the time-decayed average semantic content of recent news at each timestamp.

The lookback window (L) and decay rate (λ) jointly determine the temporal dynamics of news feature aggregation. Shorter lookback windows with higher decay rates capture rapid, short-term news effects, while longer windows with lower decay rates smooth over transient fluctuations to capture more persistent trends.

Given the uncertainty regarding optimal parameter values, a grid search approach is employed across a range of parameter combinations:

Parameter	*Values tested*
Lookback Window (L)	24, 48, 72, 168, 336, 504 hours
Decay Lambda (λ)	0.01, 0.05, 0.1, 0.25, 0.5

This yields 30 parameter combinations, each producing a distinct set of time-decayed topic and embedding features. The optimal combination is selected through the model validation procedure described in Section 3.5, which evaluates downstream prediction performance across the parameter grid.

The default parameters ($L = 336$ hours, $\lambda = 0.05$) reflect a two-week lookback with moderate decay, balancing responsiveness to recent news against stability from a broader temporal context. With $\lambda = 0.05$, an article published 24 hours ago retains approximately 30% of its initial weight ($e^{-0.05\times24} \approx 0.30$), while an article from one week ago (168 hours) retains only about 0.02% ($e^{-0.05\times168} \approx 0.0002$).

3.4.5 Dimensionality Reduction of Aggregated Embeddings

The 384-dimensional time-decayed embedding vectors pose challenges for downstream modelling due to their high dimensionality and potential for overfitting, particularly given the limited sample size relative to feature dimensionality. Dimensionality reduction techniques project these high-dimensional representations into a lower-dimensional space while preserving essential structure, thereby improving computational efficiency and model generalisation.

The UMAP algorithm described in Section 2.2.2 is the primary dimensionality-reduction method used in this research project. UMAP was chosen because it offers superior

scalability to large datasets and better preservation of global structure, making it well-suited for the continuous semantic spaces produced by transformer embeddings (McInnes et al., 2020, pp. 1–2).

The UMAP configuration employs the following parameters:

- ***n_components* = 20**: Target dimensionality, reducing the 384-dimensional input by a factor of approximately 19
- ***n_neighbors* = 15**: Local neighbourhood size, controlling the balance between local and global structure preservation
- ***min_dist* = 0.1**: Minimum distance between points in the embedding, governing clustering tightness

These parameters follow the guidelines provided by McInnes et al. (2018), who recommend *n_neighbors* values between 10 and 50 to balance local and global structure preservation, and *min_dist* values between 0.0 and 0.5 to control the tightness of the resulting clusters. The choice of 20 components provides sufficient expressiveness for downstream prediction while substantially reducing the dimensionality of the feature space.

The final output is a matrix of dimension $(T \times 20)$, where each row represents the reduced embedding for a single hourly timestamp. These 20 embedding dimensions, combined with the 14 topic count features, yield 34 news-derived features per parameter combination.

3.4.6 Feature Integration & Scaling

The complete feature matrix integrates baseline features with news-derived features to produce the input representation for predictive modelling. Table XX summarises the complete feature set:

Feature Category	*Count*	*Description*
Price Features	2	Day-ahead price, spot price
Spread Features	1	Absolute spread
Load Features	1	System load
Lagged Features	4	24h and 168h lags for price and load
Temporal Features	5	Hour, day of week, day/week/month of year
Topic Features	14	Time-decayed counts per leaf topic
Embeddings Features	20	UMAP-reduced embedding dimensions

The news-derived features (34 dimensions) account for approximately 72% of the total feature space, reflecting the research focus on extracting predictive signals from textual information.

News-derived features exhibit different scales and distributions compared to baseline features. To prevent scale-dependent biases in model training, StandardScaler normalisation is applied to the news features:

$$x_{scaled} = \frac{x - \mu}{\sigma}$$

where μ and σ are the mean and standard deviation computed from the training set. Critically, scaling parameters are fitted exclusively on training data and subsequently applied to validation and test sets, preventing information leakage from future observations into the feature normalisation.

The dataset is partitioned chronologically to preserve the temporal ordering essential for time-series forecasting:

- **Training Set**: First 70% of observations
- **Validation Set**: Next 20% of observations
- **Test Set**: Final 10% of observations

This temporal split ensures that all test set observations occur strictly after all training observations, reflecting the realistic constraint that forecasting models cannot access future information. The validation set serves for hyperparameter tuning and time-decay parameter selection, while the test set provides an unbiased estimate of out-of-sample performance SOURCE.

3.5 Model Training and Optimisation

TO BE ADDED AGAIN LATER

3.6 Model Evaluation Framework

This section details the evaluation methodology employed to assess the performance of the trained models on held-out test data. Building upon the training procedures described in Section 3.5, the evaluation framework quantifies predictive accuracy through classification metrics and establishes statistical significance of performance differences between the signal-enriched and baseline models.

Out-of-Sample Evaluation

Purpose: Establish the rigorous temporal separation between training/validation and test sets, ensuring unbiased performance estimates.

- Reiterate the chronological split: 70% train / 20% validation / 10% test (first introduced in Section 3.4.6)

- Emphasise that the test set remains completely untouched during all model development, hyperparameter tuning, and feature parameter selection

- Explain the rationale for temporal splitting in time-series contexts—standard random splits would introduce look-ahead bias by allowing the model to learn from future observations (Tashman, 2000)

- Note that all preprocessing parameters (scalers, UMAP transformations) were fitted exclusively on training data to prevent information leakage

- Describe the evaluation procedure: final models (signal and baseline LightGBM classifiers) are applied to test set features to generate predictions, which are then compared against ground-truth labels

Classification Performance Metrics

- **Accuracy**: Proportion of correctly classified observations; interpretable but potentially misleading under class imbalance

  - Formula: $\text{Accuracy} = \frac{\text{TP} + \text{TN}}{N}$

- **Macro-Averaged F1 Score**: Harmonic mean of precision and recall, averaged equally across all three classes regardless of prevalence

  - Formula: $F1_{\text{macro}} = \frac{1}{K} \sum_{c=1}^{K} \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}$

  - Rationale: Ensures minority class performance receives equal weight in evaluation, critical for trading where signals across all market conditions matter (Sokolova & Lapalme, 2009)

- **Multiclass AUC-ROC**: Area under the receiver operating characteristic curve, computed using one-versus-one (OvO) averaging

- Measures the model's ability to discriminate between classes across all probability thresholds

- OvO averaging computes AUC for each pair of classes and averages the results, providing a robust multiclass extension (Hand & Till, 2001)

- **Precision per class**: Proportion of predicted positives that are true positives

- **Recall per class**: Proportion of actual positives correctly identified

- **Confusion matrix**: Full breakdown of predictions versus ground truth to identify systematic misclassification patterns

Statistical Significance Testing

Purpose: Establish whether observed performance differences between models are statistically meaningful or could arise from sampling variation.

- **Rationale**: McNemar's test (1947) is designed for comparing two classifiers on the same dataset by analysing the contingency table of correct/incorrect predictions

- **Contingency table structure**:

  - Both models correct

  - Only signal model correct

  - Only baseline model correct

  - Both models wrong

- **Test statistic**: $\chi^2 = \frac{(|b - c| - 1)^2}{b + c}$ where b = signal correct & baseline wrong, c = baseline correct & signal wrong

- **Interpretation**: A significant p-value (< 0.05) indicates the models perform differently beyond chance; the direction of improvement is determined by comparing b and c

- **Advantages over independent-sample tests**: McNemar's test accounts for the paired nature of predictions on the same instances, providing greater statistical power (Dietterich, 1998)

- **Method**: Non-parametric bootstrap resampling (Efron & Tibshirani, 1993) to estimate uncertainty around point estimates

- **Procedure**:

  1. Resample test set observations with replacement (n = 1,000 iterations)

  2. Compute metric (accuracy, F1, etc.) on each bootstrap sample

  3. Calculate percentile-based confidence intervals (2.5th and 97.5th percentiles for 95% CI)

- **Interpretation**: Non-overlapping confidence intervals between models provide additional evidence of meaningful performance differences

- **Advantages**: Distribution-free method that makes no assumptions about the underlying metric distribution

- Report point estimates alongside 95% confidence intervals

- Indicate statistical significance levels: * ($p < 0.05$), ** ($p < 0.01$), *** ($p < 0.001$)

- Present both signal and baseline model results to enable direct comparison

3.7 Backtesting Framework

LOL

This section details the backtesting methodology employed to translate model predictions into simulated trading outcomes and evaluate economic performance. While Section 3.6 assessed predictive accuracy through classification metrics, this section evaluates the practical utility of the signal-generating pipeline by simulating trading execution on held-out test data and comparing performance against benchmark strategies.

3.7.1 Trading Strategy Specification

Purpose: Define how model predictions are translated into executable trading actions and establish the benchmark strategies for comparison.

- **Action mapping**: Model predictions (Long/Neutral/Short) are translated directly into trading positions:

- **Long (+1)**: Buy at day-ahead auction, sell at intraday spot — profits when spot > day-ahead

- **Short (−1)**: Sell at day-ahead auction, buy at intraday spot — profits when spot < day-ahead

- **Neutral (0)**: No position taken — neither profit nor loss

- **Signal generation process** (two-stage pipeline):

1. XGBoost base model generates calibrated probability estimates for spread direction

2. LightGBM meta-learner refines these into final Long/Neutral/Short signals

- Note that the Neutral class serves a risk management function, allowing the model to abstain when prediction confidence is insufficient

- Baseline Model Strategy

- Uses the price-only LightGBM classifier (without news-derived features) as the baseline

- Same action mapping as the signal model

- Enables direct comparison of news feature contribution to trading performance

- Naïve Benchmark Strategy

- **Always-long strategy**: Systematically buys at day-ahead auction and sells at intraday spot for every observation

- Equivalent to setting action = +1 for all timestamps

- Represents a simple market-making approach that profits from average positive spreads but incurs losses during negative spread periods

- Provides a lower bound for model utility—if the signal model cannot outperform this naïve strategy, the predictive complexity adds no economic value

- Transaction Cost Modelling

- **Fixed costs**: Per-trade fee in EUR/MWh representing exchange fees, clearing costs, and broker commissions

- **Proportional costs**: Percentage of trade value representing bid-ask spreads and market impact

- Returns computation:

$$R_t = \begin{cases}

\text{spread}_t - c_{\text{fixed}} - c_{\%} \cdot |\text{spread}_t| & \text{if action} = +1 \\

-\text{spread}_t - c_{\text{fixed}} - c_{\%} \cdot |\text{spread}_t| & \text{if action} = -1 \\

0 & \text{if action} = 0

\end{cases}$$

- Note that transaction costs are only incurred when positions are taken (Long or Short), not for Neutral signals

3.7.2 Performance Metrics

Purpose: Define the economic performance metrics used to evaluate trading strategy outcomes.

- **Total Return**: Cumulative profit/loss over the test period in EUR/MWh

$$\text{Total Return} = \sum_{t=1}^{T} R_t$$

- **Average Return**: Mean return per period, indicating expected profit per trade

$$\bar{R} = \frac{1}{T} \sum_{t=1}^{T} R_t$$

- **Volatility**: Standard deviation of returns, measuring return dispersion

$$\sigma = \sqrt{\frac{1}{T-1} \sum_{t=1}^{T} (R_t - \bar{R})^2}$$

- **Sharpe Ratio** (annualised): Return per unit of total risk, the standard risk-adjusted performance measure (Sharpe, 1966)

$$\text{Sharpe} = \frac{\bar{R}}{\sigma} \cdot \sqrt{N_{\text{periods/year}}}$$

where $N_{\text{periods/year}} = 24 \times 365 = 8{,}760$ for hourly data

- **Sortino Ratio** (annualised): Return per unit of downside risk, penalising only negative volatility (Sortino & van der Meer, 1991)

  $$\text{Sortino} = \frac{\bar{R}}{\sigma_{\text{downside}}} \cdot \sqrt{N_{\text{periods/year}}}$$

  where σ_{downside} is computed only from negative returns

  - Rationale: Upside volatility is desirable in trading; Sortino isolates harmful downside risk

- **Maximum Drawdown**: Largest peak-to-trough decline in cumulative returns

  $$\text{MDD} = \min_t \left(\text{Cumulative}_t - \max_{s \leq t} \text{Cumulative}_s \right)$$

  - Measures worst-case capital erosion during the strategy's lifetime

  - Critical for assessing tail risk and capital requirements

- **Win Rate**: Percentage of profitable trades among all executed trades

  $$\text{Win Rate} = \frac{\text{Number of } R_t > 0}{\text{Number of } R_t \neq 0} \times 100\%$$

- **Number of Trades**: Total count of non-zero positions (Long + Short signals)

  - Lower trade counts with similar returns indicate more efficient signal generation

  - Sharpe and Sortino ratios normalise for risk, enabling fair comparison

  - Maximum drawdown indicates worst-case scenarios that may be unacceptable regardless of average returns

- Equity Curve Visualisation

- Plot cumulative returns over time for all strategies

- Enables visual identification of:

  - Periods of sustained outperformance/underperformance

  - Regime changes where model effectiveness varies

  - Drawdown events and recovery patterns

- Sensitivity Analysis (Optional Extension)

- Evaluate performance under varying transaction cost assumptions

- Test robustness of strategy ranking to cost parameter changes

- Identify break-even transaction costs where model loses edge over naïve benchmark

4 Results

5 Discussion

6 Conclusion

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8 Appendix

Appendix 1: Declaration of Independence

Appendix 2: Python Code

Appendix 3: Excel Table with all Data

Appendix 4: AI …